KaammavariSangham (R) 1952
K.S. INSTITUTE OF TECHNOLOGY
An Autonomous Institution under VTU, Approved by AICTE, New Delhi
Accredited by NAAC with A+ Grade and NBA(CSE & ECE),
No.14, Raghuvanahalli, Kanakapura Road, Bengaluru - 560109 DEPARTMENT OF
COMPUTER SCIENCE AND DESIGN

CIE – IA 2

Degree	**:**	**B.E**	**Semester**	**:**	**VII**
Branch - Stream	**:**	**CSE//CCE/CSD**	**Course Type / Code**	**:**	**PCC / BCS703**
Course Title	**:**	CRYPTOGRAPHY & NETWORK SECURITY	**Date**	**:**	**30.09.2026**

S.No	Questions
	MODULE 2
1	Explan the Diffie-Hellamn Key exchange algorithm with an example and diagram
2	Briefly explain Elliptic Curve crytpgraphy algorithm and mention two applications.
3	Algorithm for Analog of Diffie–Hellman Key Exchange
4	Users A and B use the Diffie-Hellman key exchange technique with a common prime q = 71 and a primitive root α = 7. a. If user A has private key $X_A = 5$, what is A's public key Y_A? b. If user B has private key $X_B = 12$, what is B's public key Y_B? c. What is the shared secret key?
5	Alice and Bob use the Diffie-Hellman key exchange technique with a common prime q = 157 and a primitive root α = 5 i) If Alice has private key $X_A = 15$, Find her public key Y_A? ii) If Bob has private key $X_B = 27$, Find his public key Y_B? iii) What is the shared secret key between Alice and Bob?
6	Let q=353 and g=3. X_a=97, X_b=233. Use Diffie Hellman Key exchange algorithm to find Y_a, Y_b and Secret key K.
7	Let q=719 and g=5. X_a=157, X_b=293. Use Diffie Hellman Key exchange algorithm to find Ya, Yb and Secret key K
8	Draw the neat diagram with Diffie–Hellman algorithm with the concept of Man-in-the-Middle Attack
9	Explain the concepts of Eliptic Curve Encryption/decryption
10	Explain the analogy of Diffie–Hellman exchange algorithm in ECC.
	MODULE 3
1	Apply the concept of a digital signature to authenticate a message with a suitable simplified example
2	Explain the Applications of Cryptographic Hash functions.
3	Explain two simple hash functions (XOR and rotated XOR) and discuss why they are insecure for Cryptographic applications

4	Explain the simplified examples of using a hash function for message authentication. Describe the role of the hash value at both the sender and receiver sides.
5	Categorize the major Applications of Cryptographic Hash functions and explain the security purpose of each category.
6	Explain the use of Control Vector Encryption and Decryption for controlling key usage. Mention its advantages. With neat diagram.
7	Inspect the procedure for Symmetric Key Distribution Using Asymmetric Encryption, discuss the general principles with a neat diagram
8	Explain Symmetric Key Distribution Using Asymmetric Encryption with a solution for man in the middle attack
9	Analyze the key distribution procedure with a process of symmetric key distribution using symmetric encryption, with a neat diagram.
	List and explain the four general categories of schemes for the distribution of public keys.
10	Describe the architecture of the diagram of PKIX Architectural Model with a neat diagram.
11	Explain essential components of public key infrastructure
12	Explain major issues with KDC
13	Explain the general elements of an X.509 Certificates Construct and label the X.509 digital certificate format and illustrate the purpose of its major fields.
14	Describe the certification path for forward and reverse certificates of X.509
15	How is an X.509 certificate revoked?
16	Explain the concepts of Key and Policy Information extensions used in X.509 Certificates
MODULE 4	
1	Differentiate between Kerberos Version 4 and Version 5
2	Explain Remote User-Authentication Using Asymmetric Encryption
3	What is Kerberos? Give the summary of Kerberos version 4 message exchanger.
4	Give the scenario of mutual authentication with respect to Remote User-Authentication Using Asymmetric Encryption
5	Explain the Kerberos Version 4 diagram with message Exchanges.
6	Explain Kerberos Authentication Service with version 4 Dialogue.
7	Construct a suitable diagram to represent the overview of Kerberos authentication
8	Apply the Kerberos authentication procedure and construct a neat, labelled diagram showing the Kerberos exchanges.
9	Explain Remote User-Authentication Principles
10	Examine the NIST SP 800-63-2 E-Authentication Architectural Model with a neat labelled diagram.
11	Explain the following 1. Means of Authentication 2. Mutual Authentication and One-Way Authentication
12	Draw the diagram for Kerberos Exchanges
13	With neat diagram, Explain how request for service in another Realm